Contribution Summary

Unit code: EAT40006
Unit name: Engineering Technology Project B (ENG/CS)-H1
Submission date: 28/09/2026
Contribution for: Sprint 5

Rumeth Gunasinghe
104498227

ACKNOWLEDGMENT OF COUNTRY

[Add a statement on acknowledgment of the country.]

I identify:

the Traditional Owners of the land they lived on while completing this work (if living in Australia)

1. CONTRIBUTION EVIDENCE AND IMPACT

Contribution 1 – Project Planning and Sprint Management

Situation

- At the beginning of the sprint, the project team needed to establish a clear project direction, define responsibilities, and create a realistic plan for developing and evaluating a humanoid robot balancing system. Since the project had only two active members, effective planning was critical to ensure progress could be maintained throughout the semester.

Task

- I was responsible for assisting in the preparation of project planning artefacts, defining sprint objectives, establishing the Scrum workflow, and ensuring that project activities aligned with the requirements provided by the supervisor and client.

Action

- I created and maintained the Jira Scrum board used by the team to organise tasks, assign responsibilities, monitor progress, and track sprint completion. I participated in developing the project plan, sprint backlog, and work breakdown structure. I also assisted in preparing the project overview document and ensured project objectives were aligned with the research goal of improving humanoid robot stability through predictive control and safety filtering.
- Throughout the sprint, I regularly updated task statuses, monitored upcoming deadlines, and coordinated activities with my teammate. I reviewed project requirements and translated them into technical milestones that could be implemented incrementally.

Result

- The project obtained a structured development process with clear deliverables and measurable sprint goals. The Jira board provided visibility of task progress and allowed the supervisor to monitor development activities. This planning foundation enabled subsequent technical work to proceed systematically.

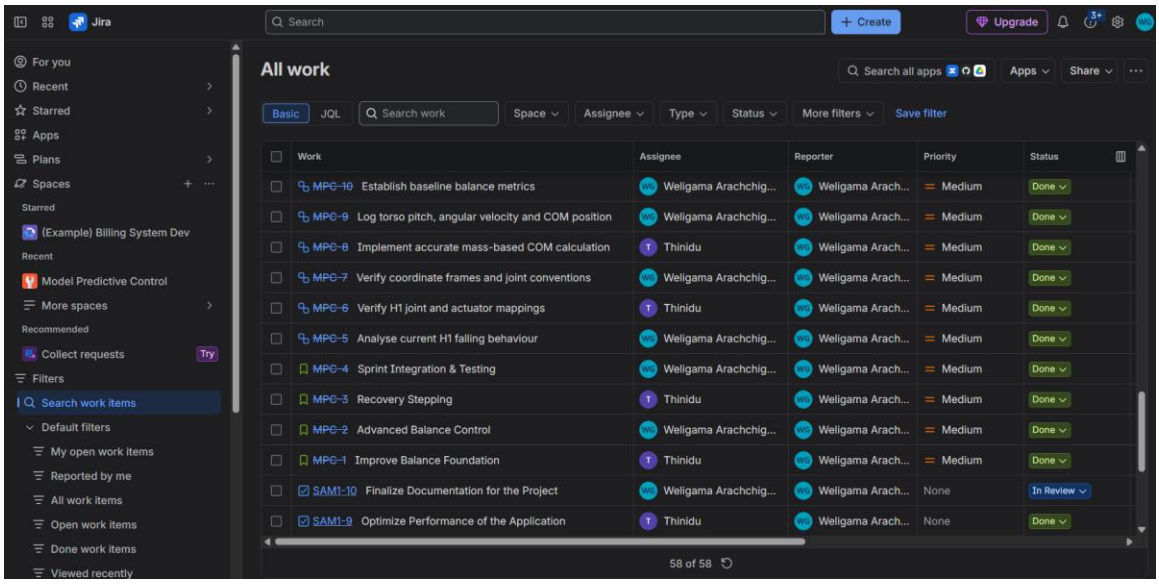
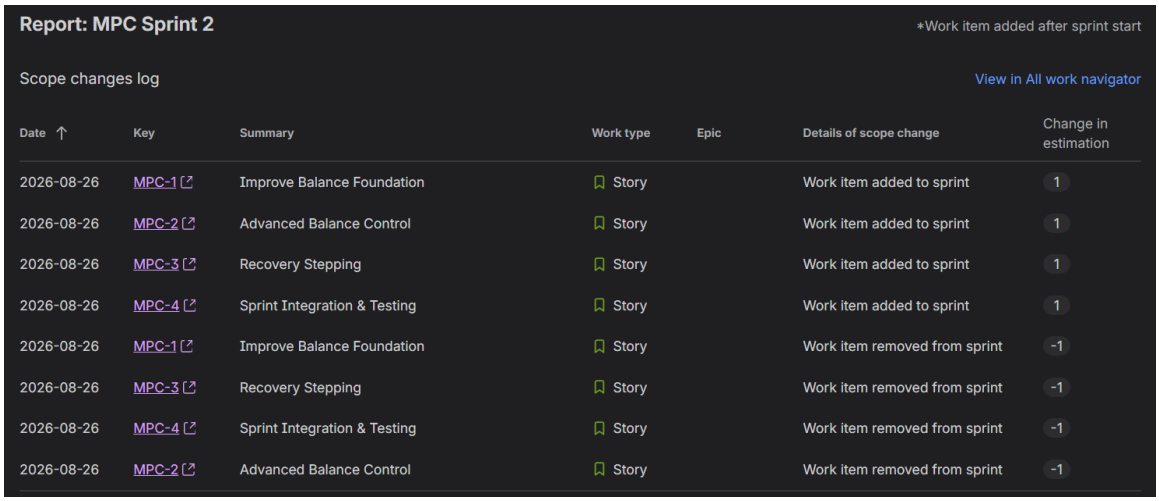


Figure 1 : sprint worklog showing tasks reported and assigned by Weligama Rumeth



Report: MPC Sprint 2

*Work item added after sprint start

Scope changes log

View in All work navigator

Date	Key	Summary	Work type	Epic	Details of scope change	Change in estimation
2026-08-26	MPC-1	Improve Balance Foundation	Story		Work item added to sprint	1
2026-08-26	MPC-2	Advanced Balance Control	Story		Work item added to sprint	1
2026-08-26	MPC-3	Recovery Stepping	Story		Work item added to sprint	1
2026-08-26	MPC-4	Sprint Integration & Testing	Story		Work item added to sprint	1
2026-08-26	MPC-1	Improve Balance Foundation	Story		Work item removed from sprint	-1
2026-08-26	MPC-3	Recovery Stepping	Story		Work item removed from sprint	-1
2026-08-26	MPC-4	Sprint Integration & Testing	Story		Work item removed from sprint	-1
2026-08-26	MPC-2	Advanced Balance Control	Story		Work item removed from sprint	-1

Figure 2 : work items added to the sprint by my to maange the project between members

Contribution 2 – Literature Review and Research Report Development

Situation

- The project required a comprehensive understanding of existing humanoid robot control methods before implementation could begin. Relevant research needed to be reviewed to justify the proposed architecture and identify suitable algorithms.

Task

- I was responsible for conducting literature research, analysing previous studies, and contributing to the preparation of the technical research report.

Action

I reviewed academic papers relating to:

- Whole-Body Model Predictive Control (MPC)
- Humanoid robot balancing
- Capture Point Control
- Linear Inverted Pendulum Models (LIPM)
- Control Barrier Functions (CBFs)
- Safe Reinforcement Learning
- Dynamic locomotion and disturbance rejection
- I analysed the advantages and limitations of different control approaches and summarised findings for inclusion within the literature review. I contributed to discussions regarding the selection of MPC combined with safety filtering as the project's primary control architecture.
- I also assisted in formatting and refining the research report to ensure academic quality and consistency.

Result

- The literature review provided a strong theoretical foundation for the project and justified the proposed control architecture. The report established a clear relationship between existing research and the project objectives.

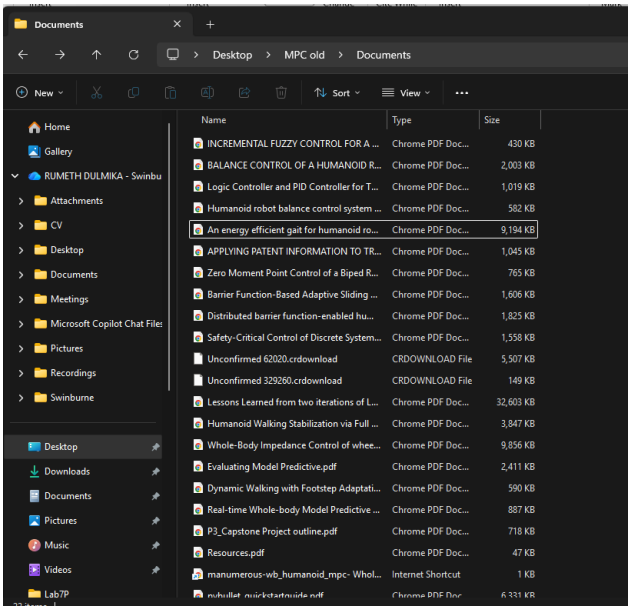


Figure 3 : Research and documents

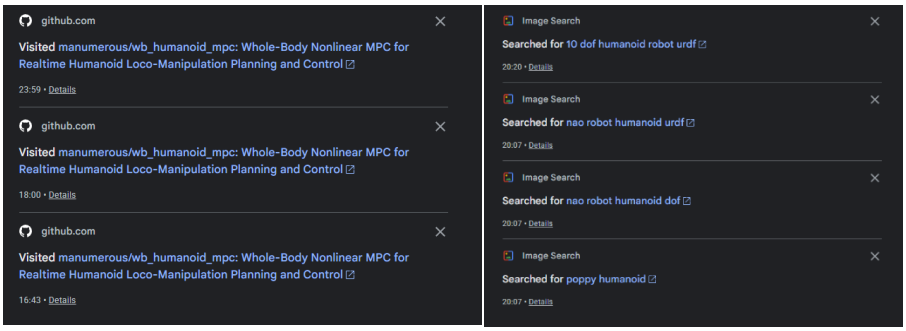


Figure 4 : research history links

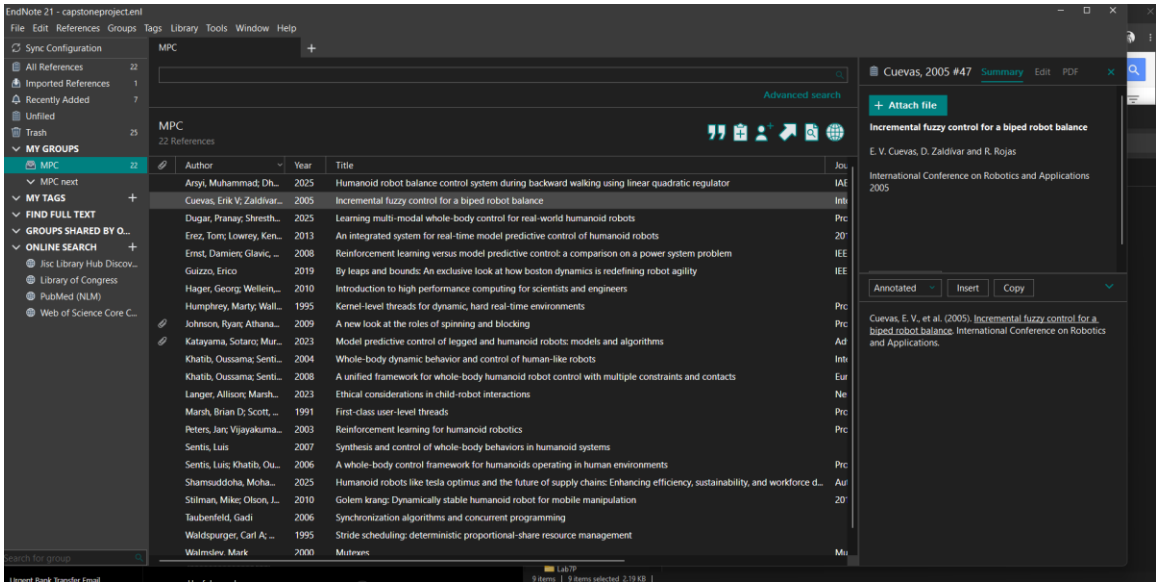


Figure 5 : reseach references

- **Estimated Time:** 20 hours

Contribution 3 – Humanoid Simulation Environment Development

Situation

- The project required a simulation platform capable of testing balancing algorithms before implementation on physical hardware.

Task

- I was responsible for assisting in the development and testing of the MuJoCo simulation environment using the Unitree H1 humanoid model.

Action

- I configured the software environment, installed required dependencies, and assisted with the integration of:
- MuJoCo physics engine
- Unitree H1 humanoid model
- Whole-body control framework
- Logging and visualisation systems
- I participated in testing simulation stability and identifying causes of controller failures. Multiple simulation runs were conducted to evaluate robot behaviour under different conditions.
- I helped verify system operation by analysing joint behaviour, centre-of-mass motion, and controller outputs.

Result

- The simulation environment became capable of executing balancing experiments and recording performance metrics required for evaluation.

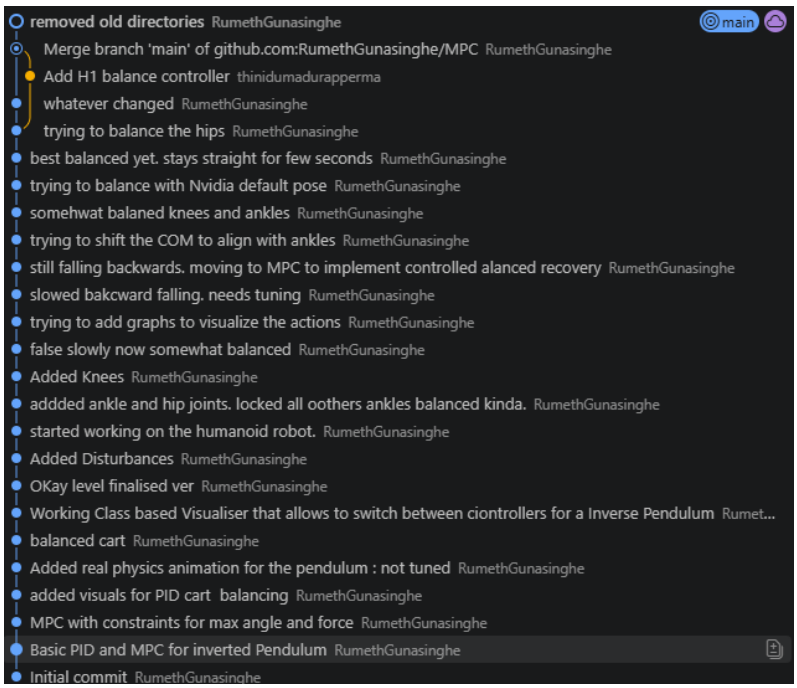


Figure 6 : github Commits that have been done by me

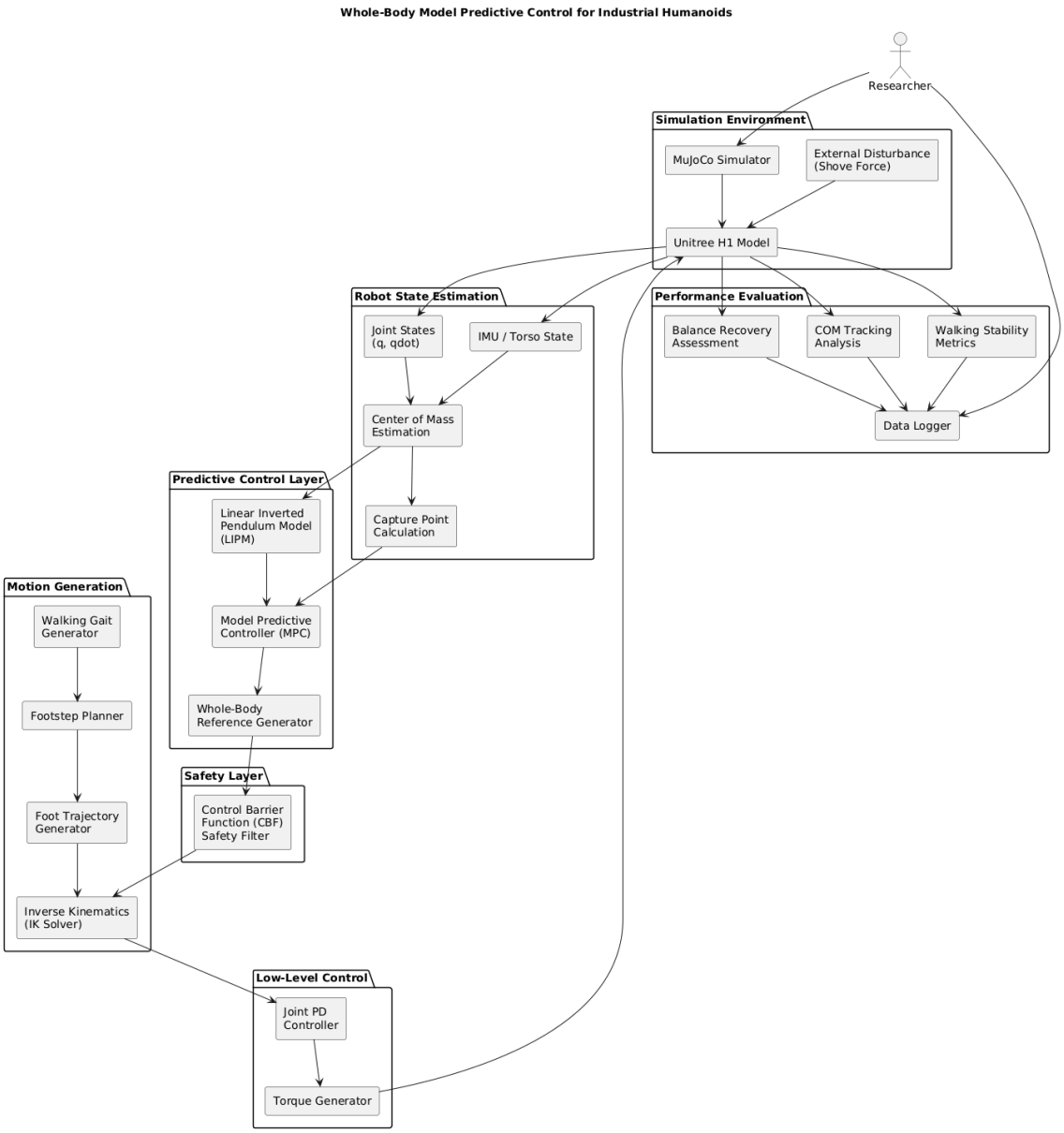


Figure 7 : MPC engine architecture diagram designed by me

- **Estimated Time:** 15 hours

Contribution 4 – Controller Debugging and Stability Investigation

Situation

- Initial simulation results showed the humanoid robot collapsing immediately after startup, preventing meaningful evaluation.

Task

- I was responsible for investigating stability issues and identifying factors contributing to robot failure.

Action

- I analysed simulation logs, joint trajectories, and centre-of-mass data to determine why the robot could not maintain balance. Through repeated testing and review of control outputs, I identified several critical issues:

    a. The safety filter incorrectly interpreted gravity-support torques as destabilising accelerations.
    b. The inverse-kinematics stage generated unrealistic joint commands.
    c. The controller update frequency was insufficient for stable balance control.

- I documented these observations and collaborated on implementing corrections. After modifications, I conducted further testing and analysed the resulting performance improvements.

Result

- The humanoid robot was able to remain standing under nominal conditions and successfully withstand an 80 N external disturbance. This represented a significant improvement compared to earlier versions that collapsed immediately.

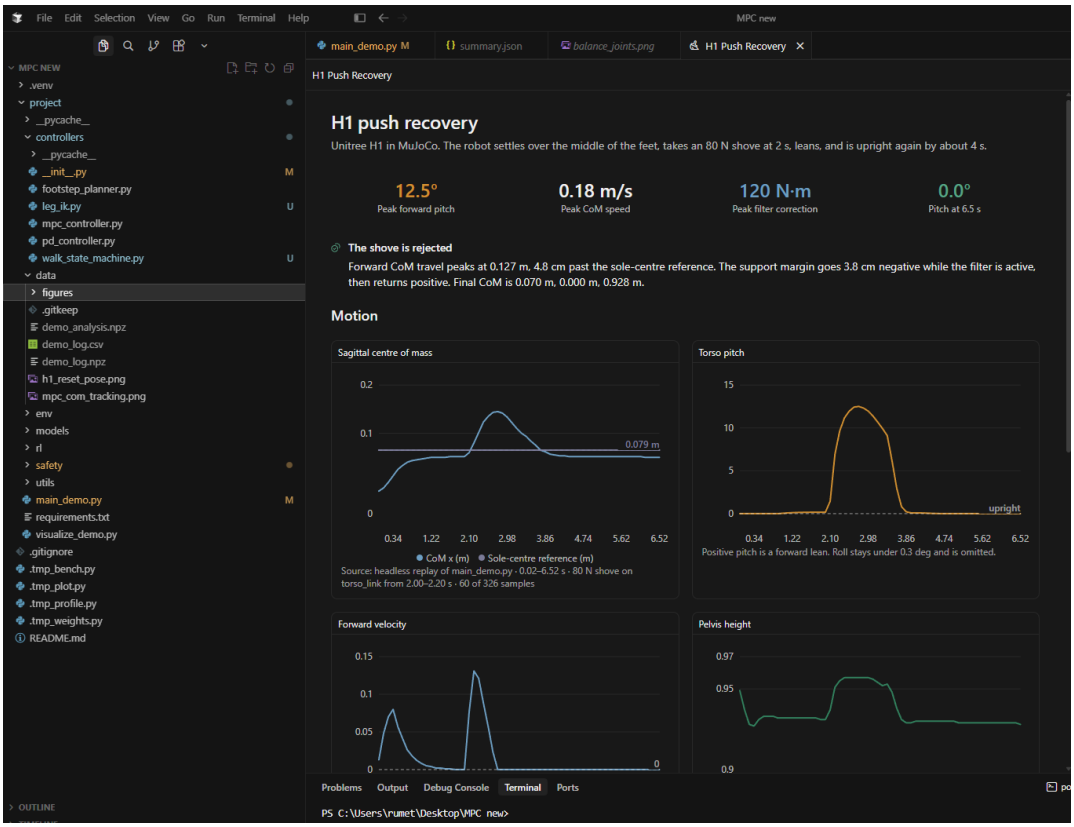


Figure 8 :Controller Debugging and Stability

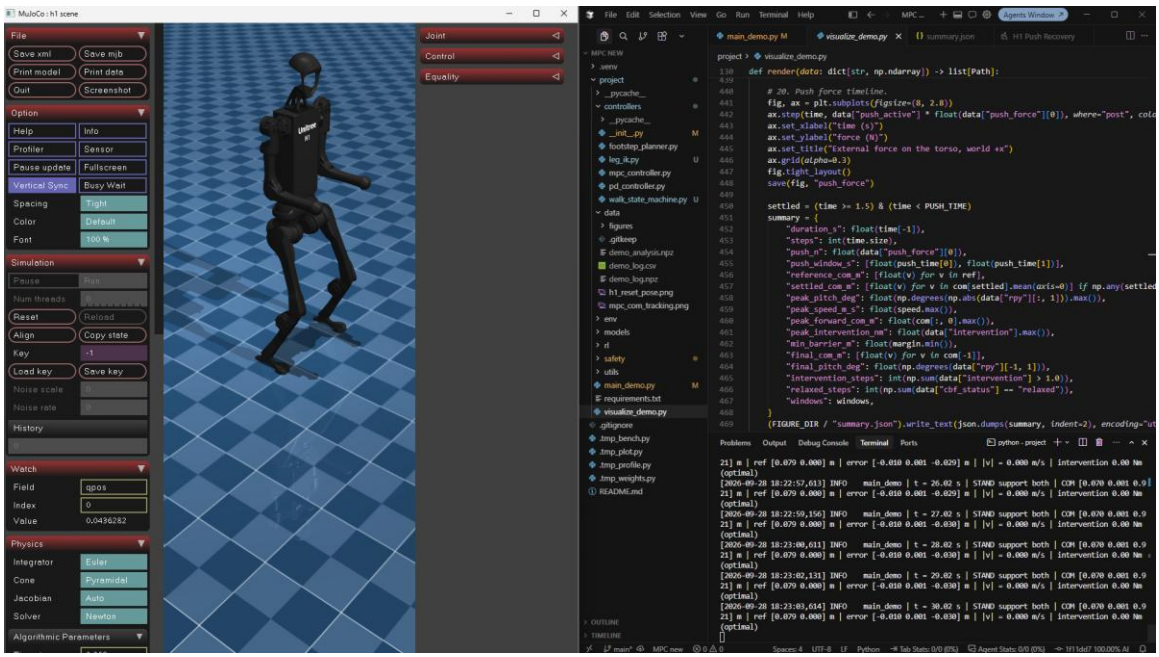


Figure 9 : Visualizing stats for debugging

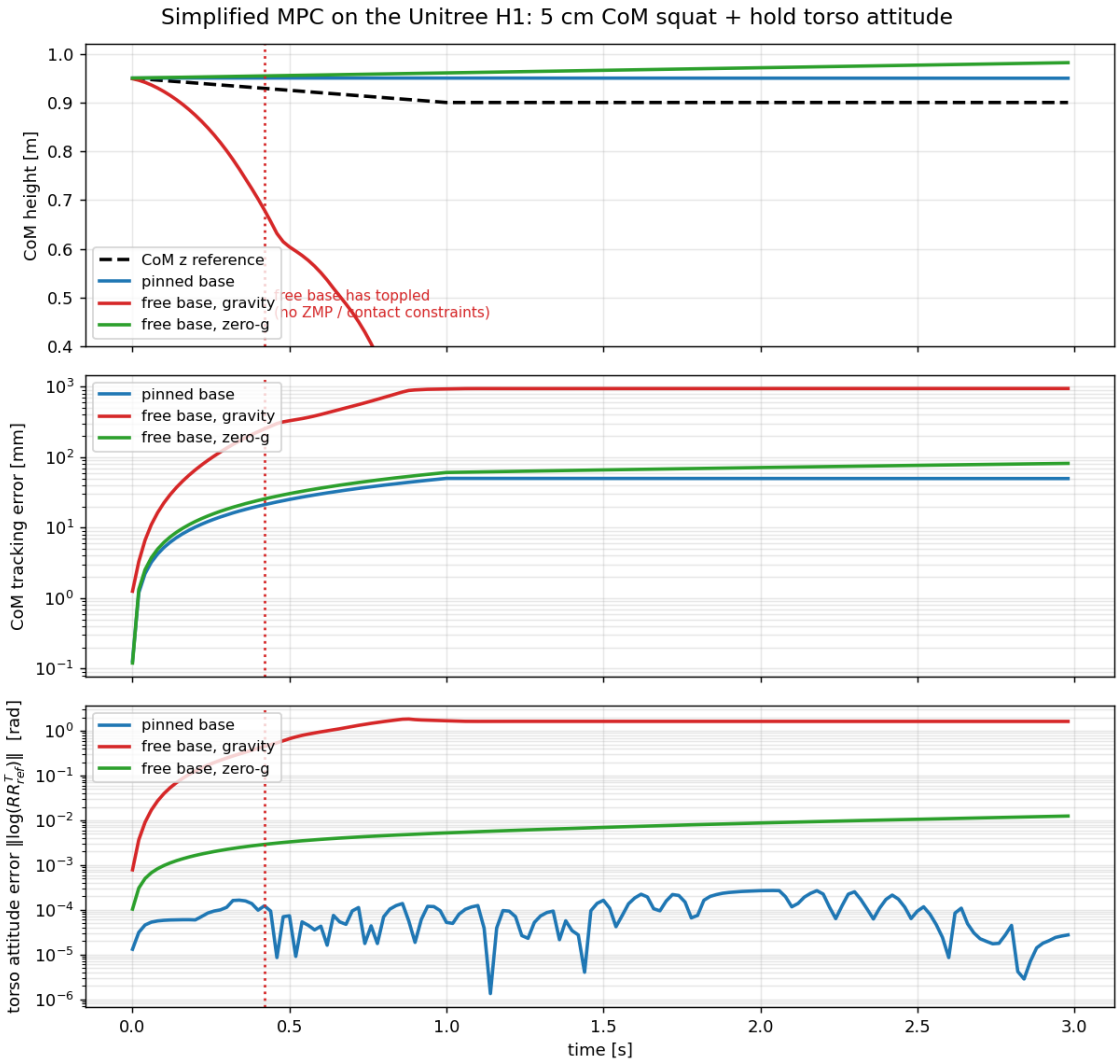


Figure 10 : Torso holding altitude when trying to shift com for walking forward

Contribution 5 – Performance Evaluation and Data Analysis

Situation

- The project required quantitative evidence demonstrating the effectiveness of the controller.

Task

- I was responsible for analysing simulation outputs and interpreting performance metrics.

Action

- I generated and reviewed performance plots including:
    - Centre-of-mass position
    - Centre-of-mass velocity
    - Centre-of-mass tracking error
    - Support-box barrier values
    - Safety-filter interventions
    - Joint tracking performance
    - Phase portraits
    - External force responses
    - Actuated joint velocity norms
- I interpreted these results to determine controller behaviour before, during, and after disturbances. Key findings included:
    - Stable CoM height around 0.93–0.95 m
    - Successful recovery from an 80 N push
    - Safety filter activation only when the support boundary was approached
    - Return to near-reference posture after disturbance rejection

Result

- The evaluation produced measurable evidence supporting the effectiveness of the balancing strategy and provided a basis for future comparison with advanced locomotion methods.

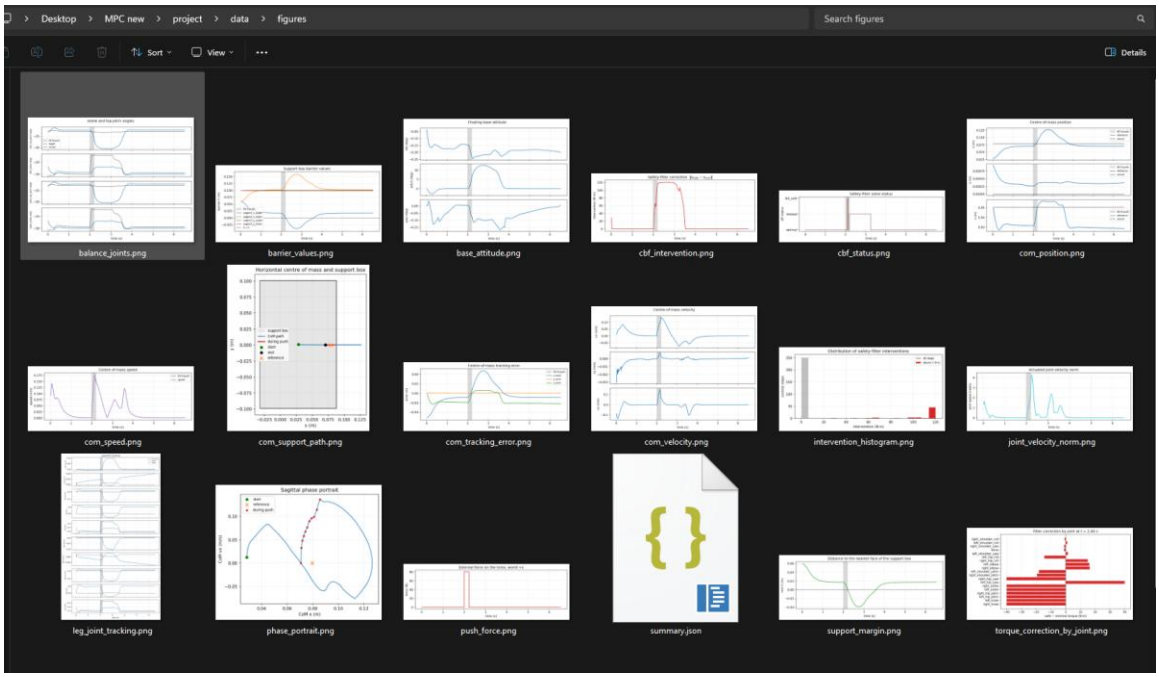


Figure 11 : graphs that were generated for testing and verification processes

- **Estimated Time:** 18 hours

Contribution 6 – Client and Supervisor Communication

Situation

- The project required regular communication with supervisors and external stakeholders to ensure expectations were met.

Task

- I was responsible for coordinating communication activities and ensuring project progress was reported.

Action

- I prepared emails to:
    - Arrange client meetings
    - Request project feedback
    - Provide project updates
    - Coordinate sprint reviews
    - Respond to supervisor concerns
- I participated in discussions regarding project scope, implementation progress, and future development priorities.

Result

- Communication channels remained active and project stakeholders received regular updates regarding progress and challenges.

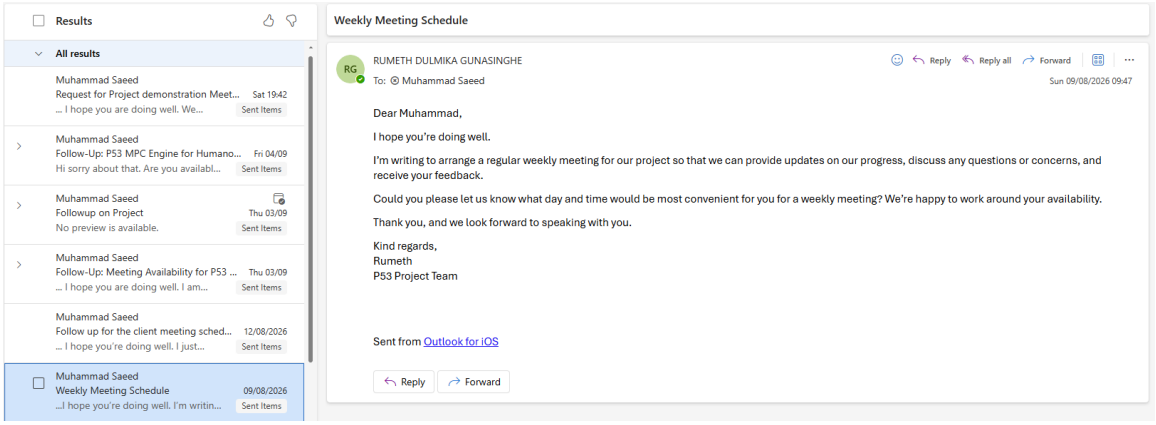


Figure 12 : Communication attempts with the client to organize a meeting

Meeting	Date	Time	Attendees	Detailed Agenda & Discussion	Demonstration / Evidence Presented	Supervisor Feedback	Improvements Required	Actions & Outcomes
1	05-Aug-2026	8:00 PM	Rumeth, Thinidu, Mr Naveed	Introduced project objectives for Whole-Body Model Predictive Control on the Unitree H1 humanoid. Discussed project scope, expected deliverables, research direction, client communication status, sprint structure and risk management. Reviewed challenges of humanoid balance, whole-body control and safety constraints.	Presented initial project plan, literature search strategy and proposed architecture.	Supervisor advised starting with simplified models before full whole-body control and maintaining weekly progress evidence.	Create project plan, establish Jira workflow, contact client, begin literature review.	Research direction approved. Sprint objectives defined.
2	12-Aug-2026	8:00 PM	Rumeth, Thinidu, Mr Naveed	Reviewed collected papers on MPC, LIPM, ZMP, Capture Point and humanoid balance control. Compared different control approaches and discussed suitability for industrial humanoids.	Demonstrated literature review progress and preliminary architecture diagram.	Supervisor recommended validating concepts in simulation before implementing advanced controllers.	Expand literature review and justify chosen MPC approach.	Decision made to use LIPM as an intermediate validation model.
3	19-Aug-2026	8:00 PM	Rumeth, Thinidu, Mr Naveed	Focused on MuJoCo environment setup and Unitree H1 model investigation. Discussed joint hierarchy, actuators, sensors and simulation capabilities.	Demonstrated successful loading of Unitree H1 and scene configuration.	Supervisor requested validation of standing behaviour before walking experiments.	Investigate robot states, COM behaviour and joint functions.	Simulation environment validated.
4	26-Aug-2026	8:00 PM	Rumeth, Thinidu, Mr Naveed	Reviewed standing control experiments. Analysed why the robot collapsed, investigated pelvis height, torso pitch and support behaviour.	Demonstrated standing simulation runs and failure cases.	Supervisor encouraged systematic testing of individual joints and documenting results.	Perform joint direction experiments and collect measurements.	Joint testing plan approved.
5	02-Sep-2026	8:00 PM	Rumeth, Thinidu, Mr Naveed	Discussed left hip, right hip, knee and ankle testing. Analysed influence of each joint on pitch stability and balance recovery. Reviewed system identification strategy.	Presented joint response graphs and measured robot reactions to perturbations.	Supervisor noted good progress in understanding robot dynamics.	Develop reduced-order dynamic model for predictive control.	System identification completed.
6	9-Sep-27	Cancelled						
7	16-Sep-2026	8:00 PM	Rumeth, Thinidu, Mr Naveed	Reviewed project delays caused by illness and communication difficulties. Discussed sprint progress, missed activities and recovery strategy.	Presented completed work despite schedule disruption.	Supervisor highlighted importance of attendance, communication and stakeholder updates.	Recover schedule and provide regular progress reports.	Recovery plan agreed.
8	23-Sep-27	Cancelled						
9	30-Sep-2026	8:00 PM	Rumeth, Thinidu, Mr Naveed	Reviewed standing recovery improvements. Discussed safety filter, capture-point correction, COM stabilisation and disturbance rejection. Analysed why the robot still falls after large pushes and limitations of ankle-only balancing.	Demonstrated stable standing behaviour before disturbance and disturbance-recovery tests.	Supervisor stated that the standing controller forms a solid foundation and recommended progressing toward stepping recovery and dynamic walking.	Implement stepping strategy, walking MPC and foot placement control.	Sprint 5 objectives defined and report preparation approved.
10	7-Oct-26							
11	14-Oct-26							
12	21-Oct-26							

Figure 13 : Weekly supervisor meetings documented by me

- **Estimated Time:** 8 hours

Total Time Spent

Activity	Hours
Sprint Planning & Jira Management	10
Literature Review & Research Report	20
Simulation Environment Development	15
Controller Debugging & Stability Investigation	25
Performance Evaluation & Data Analysis	18
Client & Supervisor Communication	8
Total	**96 Hours**

2. TEAMWORK, COMMUNICATION AND PROFESSIONAL PRACTICE

During Sprint 5, I actively collaborated with my teammate, supervisor, and client to progress the development of the humanoid robot control system. As one of only two members in the project team, I took responsibility for coordinating technical development activities, maintaining communication regarding project progress, and ensuring that technical challenges were documented and discussed with stakeholders.

I regularly communicated with my teammate regarding implementation progress, testing results, and controller design decisions. Throughout the sprint, I shared updates on the development of the walking framework, including the creation of gait generation modules, inverse kinematics components, foot trajectory generation, Model Predictive Control (MPC) integration, and Control Barrier Function (CBF) safety filtering. Communication was maintained

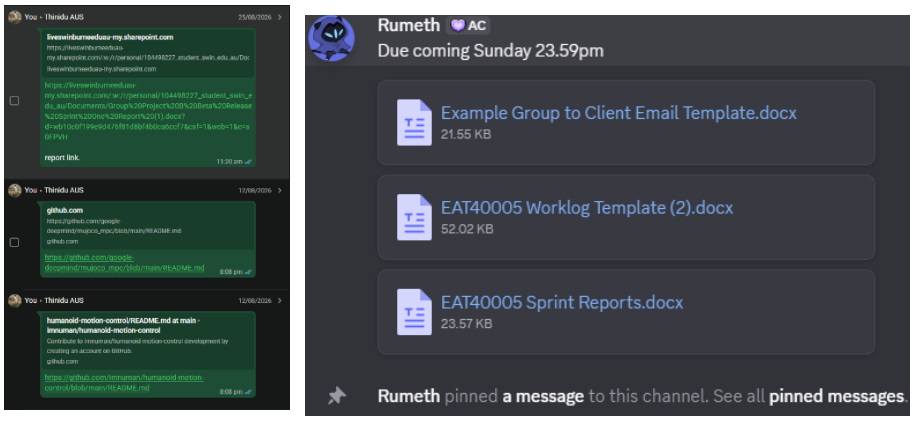


Figure 14 : communication with the team

through meetings, messages, and shared project documentation to ensure both team members remained aware of current development status and upc oming tasks.

I participated in supervisor meetings and prepared evidence demonstrating project progress. This included presenting simulation results, discussing controller performance, explaining implementation challenges, and outlining future development plans. During these discussions, I provided demonstrations of the humanoid simulation environment, including standing stability tests, disturbance rejection experiments, and walking controller integration. Evidence of these activities includes meeting records, supervisor feedback, sprint planning documents, GitHub commits, simulation screenshots, and generated plots.

Professional communication was maintained with the project client throughout the sprint. Meeting requests and progress updates were communicated respectfully and professionally. Technical progress, implementation decisions, and current limitations of the system were explained clearly to ensure that expectations regarding project outcomes remained realistic. Client communication records and email correspondence can be provided as evidence.

Collaboration with my teammate involved sharing technical findings and discussing approaches for implementing walking control on the Unitree H1 humanoid. During development, several controller failures occurred, including instability caused by incorrect inverse kinematics outputs, safety filter behaviour that unintentionally removed gravity-support torques, and timing issues between the MPC controller and physics simulation. Rather than assigning blame, these issues were analysed collaboratively and resolved through discussion, testing, and iterative improvement. This demonstrated respect for different viewpoints and a willingness to consider alternative solutions.

Responsiveness was maintained throughout the sprint by regularly updating project stakeholders regarding implementation progress and challenges. When issues were identified, updates were provided promptly so that supervisors and teammates remained informed of risks affecting project timelines. Evidence supporting this includes meeting attendance records, email correspondence, GitHub commit history, Jira task updates, and sprint planning documentation.

Overall, I believe my engagement during Sprint 5 demonstrated professional behaviour, effective collaboration, consistent communication, and a commitment to achieving project objectives despite significant technical challenges associated with humanoid robot control development.

3. REFLECTION, LEARNING AND CONTINUOUS IMPROVEMENT

Sprint 5 represented a significant transition from developing a static balancing controller toward implementing a more advanced walking-capable humanoid control framework. Although the robot did not achieve stable walking by the end of the sprint, substantial technical progress was made and several important lessons were learned regarding humanoid locomotion, control system integration, and safety-critical robotics development.

My primary contribution during this sprint was the design and implementation of the initial walking framework. This included developing gait phase management, foot trajectory

generation, inverse kinematics calculations, MPC integration, and modifications to the safety filter architecture. The objective was to move beyond standing balance and establish the software foundation required for future walking control development.

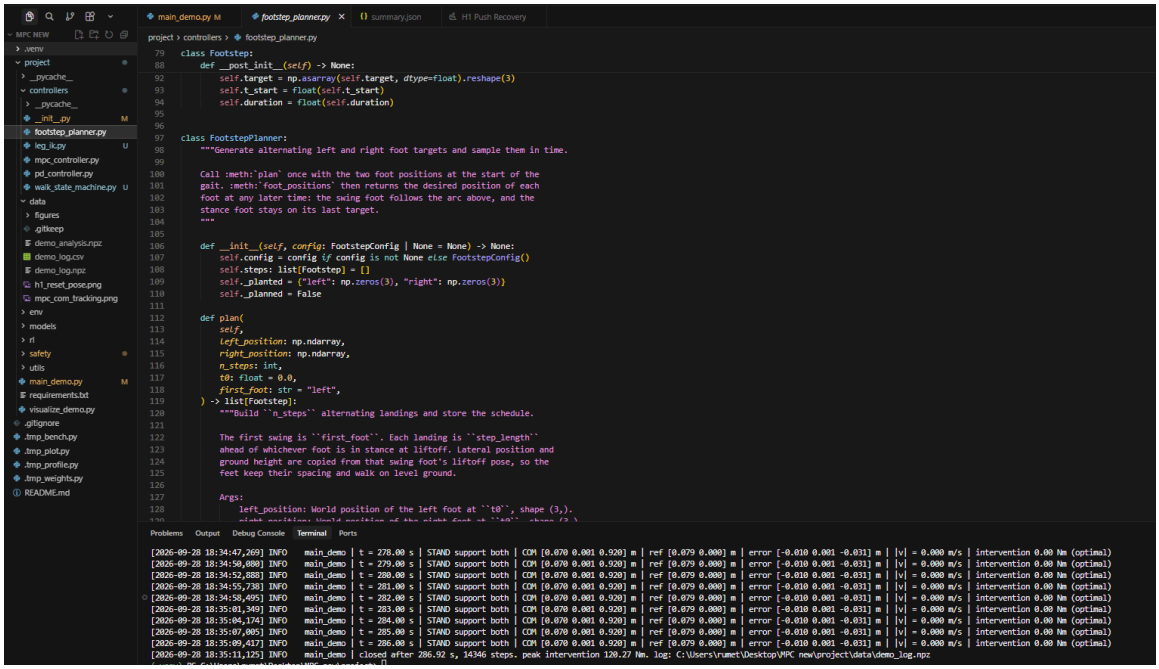


Figure 15 : Footstep planner code

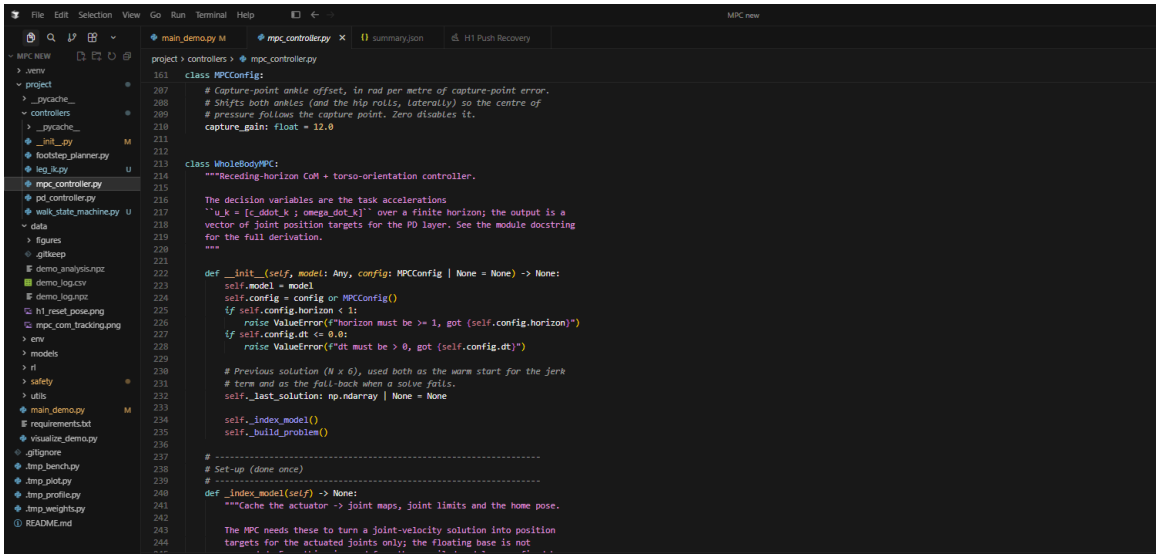


Figure 16 : MPC controller program

One of the major challenges encountered was that the robot repeatedly fell during walking experiments despite successful operation of the standing controller. Initially, it appeared that the walking controller itself was unstable. However, detailed investigation revealed that

multiple interacting subsystems were contributing to failure. The Control Barrier Function safety filter was incorrectly interpreting gravity-support torques as centre-of-mass acceleration, causing the controller to remove torques that were essential for maintaining posture. This resulted in immediate instability and collapse. Through analysis of controller outputs and simulation logs, the safety filter was redesigned to account for gravity compensation before applying corrective actions.

A second challenge involved the inverse kinematics implementation. Early versions of the IK solver generated joint targets that were physically inconsistent with the robot's kinematic structure. In particular, torso pitch corrections were being amplified into joint motions that the leg mechanisms could not realistically achieve. This caused excessive knee movement, oscillatory behaviour, and rapid loss of balance. Through repeated testing and inspection of joint trajectories, these problematic components were removed, resulting in more stable leg motion and improved posture maintenance.

Another significant lesson involved controller timing. The initial walking controller updated at only 50 Hz while the MuJoCo physics simulation operated at a substantially higher frequency. This mismatch resulted in delayed control actions and poor disturbance rejection. After analysing simulation behaviour, controller updates were moved to every physics step, greatly improving responsiveness and reducing instability during standing tests.

A key learning outcome from this sprint was the realization that successful humanoid locomotion cannot be achieved by simply combining independent control modules. Standing balance, walking control, inverse kinematics, MPC optimisation, safety constraints, and simulation timing all interact closely. A modification in one subsystem can significantly affect overall robot behaviour. This highlighted the importance of systems engineering principles when developing complex robotic platforms.

Although stable walking was not achieved, the sprint successfully established the software architecture required for future development. The robot can now maintain standing posture for extended periods, withstand small disturbances, and execute walking-related trajectory generation functions. Furthermore, the project now includes a modular walking framework consisting of gait scheduling, foot trajectory planning, inverse kinematics, MPC control, and safety filtering components.

Moving forward, several improvements have been identified. First, the current walking implementation relies on predefined swing trajectories rather than dynamically planned footsteps. Future work will focus on implementing capture-point based footstep planning so that the robot can actively place its feet to prevent falls. Second, the current reduced-order MPC model must be extended toward whole-body control capable of coordinating upper and lower body motion simultaneously. Third, controller parameter tuning should be performed using systematic optimisation methods rather than manual tuning. Finally, more comprehensive disturbance testing should be conducted to quantify robustness under different push forces and directions.

Overall, Sprint 5 provided valuable experience in humanoid robotics, Model Predictive Control, inverse kinematics, safety-critical control systems, and simulation-based validation. The challenges encountered improved my understanding of the complexities involved in whole-body humanoid control and established a strong foundation for the remaining stages of the project.